BIT 328/COM 323 OOAD ASSIGNMENT IN GROUPS OF 8 STUDENTS

USE A UNIFIED MODELLING LANGUAGE

GROUP A, C

Group Assignment: UML Modeling of a Campus Hostel Management System

Your group is required to analyze and model a **Campus Hostel Management System** used in a Karatina university to manage room allocation, student check-in/check-out, maintenance requests, and payments.

Scenario

The university provides several hostels for students. Each hostel has multiple rooms with different capacities (single, double, triple). Students apply for hostel accommodation online, are allocated rooms based on availability and priority (year of study, special needs, fee status), and must check in at the start of the semester and check out at the end. Maintenance issues (e.g., broken lights, leaking taps) are logged by students and handled by hostel staff. Payments are recorded via the university finance system, and unpaid balances can lead to denial of hostel allocation in the next semester.

Required

Working in groups of 8 students, **design a coherent set of UML models** for this system.

Your work must include at least the following:

1. **Use Case Model**
    - Identify at least **four main actors** (e.g., Student, Hostel Warden, Maintenance Staff, Finance Officer, System Administrator).
    - Develop a **Use Case Diagram** showing:
        - Main functional use cases (e.g., Apply for Hostel, Allocate Room, Check In, Check Out, Submit Maintenance Request, Approve Maintenance, Record Payment).
        - At least **two «include»** and **two «extend»** relationships.
        - Provide **brief textual descriptions** for at least **five** key use cases.
2. **Class Diagram (Structural Model)**
    - Identify and model at least **8–10 core classes** (e.g., Student, Hostel, Room, Allocation, MaintenanceRequest, Payment, Semester, UserAccount).
    - For each class, specify **key attributes** and **operations**.

- Show appropriate relationships:
    - Associations with multiplicities and role names.
    - At least **one generalization** (inheritance) hierarchy.
    - At least **one aggregation or composition** relationship (e.g., Hostel–Room).

3. **Interaction Diagram (Sequence or Communication)**
    - Choose **one critical scenario**, for example:
        - "Student applies for hostel and receives room allocation" **or**
        - "Student submits maintenance request and issue is resolved."
    - Draw a **Sequence Diagram** (or Communication Diagram) showing:
        - Boundary, control, and entity objects.
        - The key messages in order, including at least one decision/alternative fragment (alt).
4. **Activity Diagram**
    - Draw an **Activity Diagram** for the end-to-end process of:
        - "Student Check-In and Room Allocation" **or**
        - "Maintenance Request Handling."
    - Include decisions, merges, and (where reasonable) parallel activities.
5. **Short Design Rationale (1–2 pages)**
    - Explain **why** you chose particular classes and relationships.
    - Justify at least **two design decisions** (e.g., your inheritance choice, why composition vs aggregation, how responsibilities are distributed).
    - Reflect briefly on how the different UML diagrams complement each other to give a complete picture of the system.

Deliverables

- One **group report** containing all diagrams and explanations.
- All UML diagrams must be **clearly labeled**, readable, and use **standard UML notation**.

- Indicate **group members' names and registration numbers** on the cover page.

Group Assignment: UML Modeling of a Library Management System

GROUP B, E

Your group is required to analyze and design UML models for a **Library Management System** used in a university or public library to manage books, members, borrowing, returns, reservations, fines, and librarian operations.

Scenario

A university library serves students, lecturers, and librarians. The system allows members to search for books, borrow available copies, return borrowed books, renew loans, and reserve books that are currently unavailable. Librarians register new members, add new books and copies, approve borrowing and returns, and calculate fines for overdue books. The system also keeps records of book availability, due dates, reservations, and penalties. Administrators may generate reports on issued books, overdue materials, popular books, and active members.

Required

Working in groups of 8 students, **design a coherent set of UML models** for this system.

Your work must include at least the following:

1. Use Case Model

- Identify at least **four main actors** such as:
    - Member/Student
    - Librarian
    - Administrator
    - Lecturer (specialized member)
- Develop a **Use Case Diagram** showing:
    - Main use cases such as Search Book, Borrow Book, Return Book, Renew Loan, Reserve Book, Register Member, Add Book, Generate Report, Calculate Fine
    - At least **two «include»** relationships

    - At least **two «extend»** relationships
- Provide **brief textual descriptions** for at least **five** important use cases.

2. Class Diagram (Structural Model)

- Identify and model at least **8–10 core classes**, for example:
    - Book
    - BookCopy
    - Member
    - Librarian
    - Loan
    - Reservation
    - Fine
    - LibraryCatalog
    - Report
    - UserAccount
- For each class, specify:
    - Key attributes
    - Main operations/methods
- Show appropriate relationships:
    - Associations with multiplicities and role names
    - At least **one inheritance/generalization** relationship
    - At least **one aggregation or composition** relationship

3. Interaction Diagram (Sequence or Communication)

- Choose **one major scenario**, for example:
    - "A member borrows a book successfully"
    - "A member reserves an unavailable book"
    - "A librarian processes a returned overdue book and fine"
- Draw a **Sequence Diagram** or **Communication Diagram** showing:

- Boundary, control, and entity objects
- The messages exchanged in correct order
- At least one conditional/alternative flow using **alt**

4. Activity Diagram

- Draw an **Activity Diagram** for one of the following processes:
  - Borrowing a book
  - Returning a book
  - Reserving an unavailable book
- Your activity diagram should include:
  - Initial node and final node
  - Action states
  - Decision points
  - Merges and, where appropriate, parallel activities

5. Short Design Rationale (1–2 pages)

Explain:

- Why you selected the classes used in your model
- Why specific relationships were chosen
- At least **two important design decisions**, for example:
  - Why Book and BookCopy are modeled separately
  - Why Fine is associated with Loan
  - Why Member may be generalized into Student and Lecturer
- How the UML diagrams complement each other in describing the whole system

Deliverables

- One **group report**
- All UML diagrams must be **well labeled**, neat, and use **standard UML notation**
- Include **group members' names and registration numbers**
- State any assumptions made in the modeling process

GROUP D, F

Group Assignment: UML Modeling of a Hospital Management System

Your group is required to analyze and design UML models for a **Hospital Management System** used to manage patient registration, appointments, consultations, admissions, billing, laboratory requests, and discharge processes.

Scenario

A hospital serves patients through several departments such as reception, consultation, laboratory, pharmacy, billing, and inpatient wards. Patients register at reception, book appointments, consult doctors, receive prescriptions or lab test requests, make payments, and may be admitted for further treatment. Doctors diagnose patients and prescribe treatment, nurses monitor admitted patients, laboratory staff process tests, and administrators generate operational reports.

Task

Working in groups of 8 students, **design a coherent set of UML models** for this system.

1. Use Case Model

- Identify at least **four main actors**, for example:
    - Patient
    - Doctor
    - Receptionist
    - Nurse
    - Laboratory Technician
    - Administrator
- Develop a **Use Case Diagram** showing:
    - Main use cases such as Register Patient, Book Appointment, Consult Patient, Request Lab Test, Record Diagnosis, Prescribe Medication, Admit Patient, Discharge Patient, Process Payment, Generate Report.
    - At least **two «include»** relationships
    - At least **two «extend»** relationships
- Provide **brief textual descriptions** for at least **five** important use cases.

2. Class Diagram

- Identify and model at least **8–10 core classes**, for example:
    - Patient
    - Doctor
    - Appointment
    - MedicalRecord
    - Prescription
    - LaboratoryTest
    - Bill
    - Payment
    - Ward
    - Admission
- For each class, specify:
    - Key attributes
    - Main operations/methods
- Show appropriate relationships:
    - Associations with multiplicities and role names
    - At least **one inheritance/generalization** relationship
    - At least **one aggregation or composition** relationship.

3. Interaction Diagram

- Choose **one major scenario**, for example:
    - "A patient books and attends an appointment"
    - "A doctor examines a patient and requests a lab test"
    - "A patient is discharged after billing is completed"
- Draw a **Sequence Diagram** or **Communication Diagram** showing:
    - Boundary, control, and entity objects
    - Messages exchanged in the correct order

- At least one **alt** fragment for conditional flow such as payment success/failure or admission/discharge decision.

4. Activity Diagram

- Draw an **Activity Diagram** for one of the following processes:
  - Patient registration and appointment booking
  - Consultation and treatment process
  - Admission to discharge workflow
- Your activity diagram should include:
  - Initial and final nodes
  - Action states
  - Decision points
  - Merges and, where appropriate, parallel activities

5. Short Design Rationale

Write **1–2 pages** explaining:

- Why you selected the classes in your model
- Why specific relationships were chosen
- At least **two major design decisions**, for example:
  - Why MedicalRecord is associated with Patient
  - Why Admission and Ward are modeled separately
  - Why Doctor and Nurse may inherit from a common Staff class
- How the UML diagrams complement each other in describing the entire hospital system.

Deliverables

- One **group report** in PDF format
- All UML diagrams must be **well labeled**, neat, and use **standard UML notation**
- Include **group members' names and registration numbers**
- Clearly state any assumptions made in the modeling process

GROUP G, H

Group Assignment: UML Modeling of a Bank Management System

Your group is required to analyze and design UML models for a **Bank Management System** used to manage customer accounts, deposits, withdrawals, fund transfers, loan applications, and transaction records.

Scenario

A bank provides services to customers through branch staff, tellers, loan officers, and system administrators. Customers can open accounts, deposit money, withdraw money, transfer funds, check account balances, and apply for loans. Tellers' process daily cash transactions, managers approve certain operations such as large withdrawals or loans, and administrators maintain user accounts and generate reports. The system keeps records of customers, accounts, transactions, loans, and payments.

Required

Working in groups of 4–6 students, **design a coherent set of UML models** for this system.

1. Use Case Model

- Identify at least **four main actors**, for example:
    - Customer
    - Teller/Cashier
    - Manager
    - Loan Officer
    - System Administrator
- Develop a **Use Case Diagram** showing:
    - Main use cases such as Open Account, Deposit Money, Withdraw Money, Transfer Funds, Check Balance, Apply for Loan, Approve Loan, Generate Statement, Generate Report.
    - At least **two «include»** relationships
    - At least **two «extend»** relationships
- Provide **brief textual descriptions** for at least **five** important use cases.

2. Class Diagram

- Identify and model at least **8–10 core classes**, for example:
    - Customer
    - Account
    - SavingsAccount
    - CurrentAccount
    - Transaction
    - Loan
    - Payment
    - Branch
    - Employee
    - Statement
- For each class, specify:
    - Key attributes
    - Main operations/methods
- Show appropriate relationships:
    - Associations with multiplicities and role names
    - At least **one inheritance/generalization** relationship, such as Account generalized into SavingsAccount and CurrentAccount, or Employee generalized into Teller and Manager.
    - At least **one aggregation or composition** relationship

3. Interaction Diagram

- Choose **one major scenario**, for example:
    - "A customer deposits money into an account"
    - "A customer transfers funds between accounts"
    - "A customer applies for a loan and the manager approves it"
- Draw a **Sequence Diagram** or **Communication Diagram** showing:

- Boundary, control, and entity objects
- Messages exchanged in the correct sequence
- At least one **alt** fragment for cases such as insufficient funds, invalid account details, or rejected loan application.

4. Activity Diagram

- Draw an **Activity Diagram** for one of the following processes:
  - Opening a new account
  - Processing a customer withdrawal
  - Loan application and approval workflow
- Your activity diagram should include:
  - Initial and final nodes
  - Action states
  - Decision points
  - Merges and, where appropriate, parallel activities

5. Short Design Rationale

Write **1–2 pages** explaining:

- Why you selected the classes in your model
- Why specific relationships were chosen
- At least **two major design decisions**, for example:
  - Why Transaction is associated with Account
  - Why Account is specialized into SavingsAccount and CurrentAccount
  - Why Loan processing is separated from normal account transactions
- How the UML diagrams complement each other in describing the entire bank management system.

Deliverables

- One **group report**
- All UML diagrams must be **well labeled**, neat, and use **standard UML notation**
- Include **group members' names and registration numbers**
- Clearly state any assumptions made in the modeling process